
Specification-Driven Development

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How to make AI implementation reliable by treating specifications, not code, as the asset that lasts

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Preface

You already know how to write code. This book is about something you likely haven't written down yet: the decision the code is supposed to express. Two examples run through the chapters ahead, a multi-location clinic group trying to let patients book appointments online, and a retail bank trying to move a core ledger off a COBOL codebase nobody currently at the bank designed. Neither gets explained in full here. Chapter 1 opens with the clinic. Chapter 9 opens with the bank. This page only tells you they're coming.

By the last chapter, you'll be able to write a specification, a vision, a set of requirements, a use case, a domain model, and the architecture and skill conventions that surround them, precise enough that an AI assistant implements it correctly on the first pass, before review has to find what a guess got wrong. You'll also be able to run the same discipline in reverse: reading a system that was never specified, and pulling a working specification out of its behavior, because in most legacy systems, behavior is the only documentation that survived.

Some of what follows will read like process, because it is process: writing a specification asks for hours before it hands any back. The next page sorts the chapters ahead into three paths, one for an engineer writing a single feature, one for an architect keeping several teams consistent with each other, and one for a manager deciding how a team should adopt this. Pick the one that matches what you're actually deciding this week.

How to read this book

The three paths below are cuts through one argument, not three different books, and the default is to ignore them and read straight through from Chapter 1 to Chapter 11. Take a path when you have a specific decision in front of you and want the chapters that bear on it first. Nothing in a path is a substitute for the chapters it skips; it is an order of arrival.

■ PRACTITIONER

Read Chapters 1 through 11 in order. This path gives you the complete methodology, front to back, for applying it to your own feature or team.

● ARCHITECT

Read the Foreword, then Chapters 2, 4, 6, 7, and 11, then Appendix C. This path runs from vision through architecture and skills and ends at the fully worked specification template, for a reader who owns consistency across teams rather than a single feature.

◆ MANAGER

Read the Foreword, Chapter 1, Appendix A, Chapter 8, Chapter 10, and the Afterword. This path makes the case for the change and its effect on team structure and roles, without the detail of writing a specification yourself.